Ten Emerging Open Source Technologies

This article covers emerging open source technologies—the many trends that drive success today and will do so in the future. These range from how the entire tech industry is getting reshaped by digital transformation and the makeover of all tech enterprises from 'Digital Immigrants' to 'Digital Natives', to the rise of innovation accelerators and the emergence of next generation open source tech platforms.

With open source technology having been around since quite a long time now, a thriving global community has grown around it; so code is shared among developers and everyone can test, re-build and learn from each other. As industry has begun adopting open source technologies in almost all verticals, many new technologies have used open source as their very foundation.

Here are some of the areas in which groundbreaking open source technologies are set to revolutionise the world as we know it.

1. Open source in machine learning

Machine learning (ML) is the study of algorithms that use large data sets to learn, generalise and predict. The most exciting aspect of ML is that with more data, the algorithm improves its predicting power. ML has acted as a strong base for self-driving cars, speech recognition, home automation products and much more. Machine learning is closely related to computational statistics and also focuses on making predictions via computers. It is regarded as an effective method for deploying complex models and algorithms that lend themselves to prediction in the commercial space -- this is known as predictive analysis. Machine learning tasks are broadly categorised into supervised learning (semi-supervised learning, active learning and reinforcement learning) and unsupervised learning.

The most recent ML engines that have been open sourced by various IT giants are Google Cloud Machine Learning Engine, TensorFlow by Google, Amazon's ML engine for AWS, Unity ML Agents, Apache PredictionIO, Microsoft Distributed Machine Learning Toolkit, etc.

- *Google Cloud Machine Learning Engine*: This is regarded as a managed service enabling users to easily build operational ML models to work on any type or size of data. It makes TensorFlow Model perform large scale training on managed clusters, and is also equipped to manage the trained models for large scale online and batch predictions. It is integrated with Google Cloud,